



Crunchyroll is a premier global video streaming application dedicated to anime, manga, and Asian media content. It serves as an industry-standard blueprint for building high-bandwidth, real-time streaming services.

Core Functional Features

1. **Simulcast & Video Streaming:** High-definition video streaming utilising adaptive bitrate algorithms (HLS/DASH) with support for multi-language subtitles and audio tracks.
2. **Content Discovery & Search:** Dynamic filtering by genre, release year, and popularity, combined with instant search and autocomplete.
3. **Personalisation & Watchlists:** Features "Crunchylists" (custom playlists), watch history syncing, and continuous playback recovery.
4. **Offline Storage:** Secure, DRM-encrypted downloading for offline playback on mobile devices.

Technical Architecture

1. **Architecture Pattern:** MVVM (Model-View-ViewModel) with Clean Architecture and Unidirectional Data Flow (UDF).
2. **Networking & Asynchronous Stream:** Retrofit / Ktor for RESTful API calls combined with Kotlin Coroutines & Flow for reactive UI state updates.
3. **Media Playback Engine:** AndroidX Media3 / ExoPlayer for handling video streaming, HLS parsing, audio track switching, and custom controller bindings.
4. **Data Persistence:** Room DB for offline metadata/watch logs and DataStore for user session tokens and preferences.
5. **Remote Configuration:** Runtime feature flags and configuration loading to enable A/B testing and incremental feature rollouts without requiring app updates.

